

Lead Software Architect

Experienced iOS Architect specializing in health, fintech, enterprise realms and social media platforms, with a profound understanding of crafting robust solutions. A leader in optimizing app performance and modernizing legacy systems. Proficient in GraphQL integration, Swift/SwiftUI utilization, and mentoring teams. Adept at enhancing accessibility and integrating native iOS innovations.

Allow our skilled team to become an integral part of yours!

PROFESSIONAL EXPERIENCE

Twitter, Inc

Lead iOS Architect

Oct 2021– Present

Responsibilities:

- ❑ Built native iOS GraphQL client for one of the country's largest social media platforms and replaced the old REST API with the new generic, reusable and completely independent GraphQL client.
- ❑ Led a team of 5 iOS developers, fostering a collaborative environment that resulted in a 30% reduction in bug reports and a 20% increase in feature delivery.
- ❑ Optimized app performance by conducting thorough code reviews and implementing advanced memory management techniques, resulting in a 40% decrease in crashes and improved app response time by 50%.
- ❑ Collaborated with cross-functional teams (designers, product managers, and backend engineers) to define technical requirements, scope, and project timelines, delivering high-quality features on time and within budget.
- ❑ Leveraged Swift and SwiftUI to refactor legacy code, reducing technical debt by 30% and simplifying the onboarding process for new team members.
- ❑ Mentored junior developers, providing code reviews, guidance, and training sessions, resulting in their professional growth and an increase in overall team productivity.
- ❑ Leveraged the type of system by applying Phantom Type to make some of bugs and mistakes impossible to happen right at the compile time.
- ❑ Legacy Code Maintenance: Managed and supported complex legacy codebase, ensuring stability and seamless operation.
- ❑ Mobile App Support: Provided maintenance and support for iOS and Android mobile applications, resolving issues promptly.
- ❑ Thorough Testing: Ensured code reliability by conducting comprehensive self-testing before submitting for Quality Assurance.
- ❑ UI Development: Crafted user-friendly UI views from rough descriptions, enhancing user experience and product usability.
- ❑ Agile Team Player: Collaborated within Agile teams, participated in peer code reviews, and engaged in constructive design discussions.

- ❑ Applied down sampling technique at a low-level to reduce memory consumption and improve performance by 75% on an infinite scrolling view. Issue caused by misunderstanding on how UIImage class memory allocation works under the hood.
- ❑ Refactored a huge legacy codebase before getting into Unit Testing. The idea here is breaking the project into small modules via Swift Package Manager as separate packages to be tested in isolation.
- ❑ Added and used Firebreak types to wrap old API. The benefit here is allowed me to incrementally refactor the codebase without having to fix and edit every single error Xcode throws during the refactoring and building phases. This technique helped me to ship and release the app without having to wait for the whole refactoring to be complete.
- ❑ Moved Domain Logic and formatted to the Model layer of the app. Incrementally modernized the old API and added Unit Testing coverage.
- ❑ Removed Firebreak types (wrappers around old API) and incrementally proceeded with Unit Testing. • Applied Unit Testing best practices: An example here is utilizing value over reference types when classes are not necessary.
- ❑ Advanced Unit Testing techniques applied: Effectively and efficiently dealt with value and reference types to simplify Unit Testing. By separating the behavior and the logic code from any class. This significantly makes Unit Testing and writing Test Doubles much easier and pleasant to work with.
- ❑ Wrote Keychain controller (native solution) and replaced dependencies on other clients' projects.
- ❑ Creating new advanced native iOS apps from scratch based on project requirements.
- ❑ Created Python-based Django Restful API with admin backend. No third-party dependencies. Created reusable SwiftUI components and make them ready-to-use Swift Packages via the Swift Package Manager. Projects available on my page.
- ❑ Assisted with converting existing iOS apps from UIKit to SwiftUI
- ❑ Worked on complex exciting projects by implementing Core Data and iOS native data persistence solutions instead of relying on third-party libraries
- ❑ Applied Lifecycle and drip email campaigns for my SaaS clients that helped them increase their conversions by %10 in a few weeks
- ❑ Utilized UIKit's accessibility APIs to provide descriptive labels, hints, and traits for UI elements, enhancing their accessibility for Voiceover users.
- ❑ Designed and implemented dynamic type support, allowing users to adjust font sizes based on their preferences and accessibility needs.
- ❑ Developed macOS applications using AppKit and Cocoa framework, leveraging the full power of the native macOS development ecosystem.

(Reddit, Inc)

Senior iOS Engineer

April 2021 – Oct 2021

Responsibilities:

- Built completely new trending projects from scratch that should be available to tens of millions of users on iOS app.
- Refactored one of the most complex and highly coupled legacy parts in the codebase. With the goal of high cohesion, maintainability and ease of reusability in other parts of the app.

- Worked on implementing safety features requested by the company in the whole iOS App. This work was done in many different areas of the codebase.
- Rebuilt the whole User profile section from scratch. Everything is adaptive, testable, reusable and easily editable. This work is being used by 50+ million active users monthly.
- Created and integrated the Creator Tools on top of the User Profile that allowed content creators to easily create, share and get credited for their contents.
- Unit testing. Backend integration of both REST and GraphQL. Including CI CD (mobile continuous integration)
- Tough bug fixes are caused by either singleton pattern or poorly structured code, or UI related.
- Built dynamic UI components that adapt to different screen sizes automatically and totally reusable.
- Built new UI components that based on SwiftUI that are fully adaptive to any screen size. •
- Other tasks as assigned.

Modern Industrial Marketing, LLC

Senior iOS Engineer

May 2019 – May 2020

Responsibilities:

- Architecture the codebase based on the extended MVC design pattern while adhering to the best S.O.L.D principles
- Rewrote the whole data sync of the codebase from scratch by using protocols and advanced caching systems with the help of the new Codable protocol
- Converted parts of the codebase from Objective-C to Swift 4.2
- Created many reusable components for common UI modules used in other clients' apps
- Wrote and added sophisticated highly efficient table view/collection view with multiple cells used heavily in the codebase.
- Participated in the Scrum sprints and other agile methodologies activities
- Fixed crucial hard to find bugs related to Core Data and multithreading issues
- Created new and modern Web Viewer and highly optimized video player for the content viewing section of the framework
- Eliminated many of the third-party dependencies by using iOS native solutions as part of the SDK.
- Created complex calculators for the company clients on their end apps. These calculators use advanced match calculations based on Excel sheet data after parsing

Halsey Solutions, LLC

Senior iOS Developer

Nov 2018 – Apr 2019

Responsibilities:

- My goal was to make advanced integrations into the Roundware framework and create the new Transformer app based on their updated codebase.

- Integrated and implemented new features into the Roundware framework
- Designed and developed the new Transformer app from scratch based on the Roundware framework
- Architected new app based on the best coding practices that adhere to the S.O.L.D principles and replaced singleton class with dependency injection
- Used protocol-oriented networking approach when working with JSON and remote data iOS Developer

Stalwart Development Group LLC

Senior iOS Developer

May 2018 – Apr 2019

Responsibilities:

- Developed native iOS apps for the construction industry
- Analyzed and optimized UI and backend application code for efficiency and performance
- Worked with other team members to investigate other design approaches, prototype new technology, and evaluate technical feasibility
- Designed and created new Restful API and backend servers that communicate with the new apps
- Worked on creating new content management systems for the Restful API using the Django framework
- Developed custom websites and other automated tools that made maintenance more efficient and easier
- Performed search engine optimization
- Provided technical support for high-end company employees
- Performed weekly assessments of all websites and software to look for potential problems or crashes

Yaqeen Solutions

iOS Developer

Jan 2017 – May 2018

Responsibilities:

- Developed native iOS apps for the construction industry
- Lead a team of offshore developers in development of an iOS application
- Built up best practices and coding standards for the iOS department
- Extensive work with housing data, via REST services with AFNetwork.
- Developed applications based on Object and Protocol Oriented Programming standards
- Followed the Model-View-Controller (MVC) design pattern.
- Data which was meaningfully displayed on a map through MapKit.
- Unit Testing with KIF testing framework.
- Collaborated with designers and business to design and develop mobile applications.
- Rapid prototyping of concept apps for marketing and sales.

Kaad Center

iOS Developer

Jan 2014– Jan 2017

Responsibilities:

- Worked closely with product managers and designers to define a rich iOS experience for the user.
- Supported the architect in improving the native iOS codebase.
- Explored new technologies and shared findings with the team.
- Supported and streamlined company policies, procedures, mission, and standards of ethics and integrity.
- Led the initial phase of medium to large projects to come up with a high-level design and estimates.
- Developed complete media player component based on Apple's native framework with caching advanced policy.
- Fixed crucial bugs and applied the best S.O.L.D principles into the codebase.
- Improved the performance of the app with memory management best practices. Example (Battery consumption, unused resources that were allocated, caching policy, binary files and frequent fetching from File System and finally down sampling images fetched from the API.
- Participating in discussions with clients to clarify our approach and update them accordingly.
- Designing prototypes for apps

Microsoft, Inc.

C, C++ Developer

Apr 2009 – May 2013

Responsibilities:

- Developed, improved, ported and debugged numerous command-line based programs and plugins across multiple platforms using C, C++.
- Design and Develop: Contribute to the design and development of core C++ components for Microsoft's flagship software products, focusing on performance, scalability, and maintainability.
- Code Optimization: Collaborate with cross-functional teams to optimize existing C++ codebase, enhancing application performance and memory utilization to ensure a seamless user experience.
- Bug Fixing and Troubleshooting: Investigate and resolve complex technical issues by debugging and providing fixes for C++ code, ensuring high-quality and stable software releases.
- Code Reviews and Collaboration: Participate in code reviews, provide constructive feedback, and mentor junior engineers, promoting best practices and maintaining code quality standards.
- Innovation and Continuous Improvement: Stay up to date with industry trends and emerging C++ technologies, proposing innovative solutions to enhance development processes and software capabilities within Microsoft's ecosystem.

EDUCATION

Bachelor of Computer Science

- BA in Computer Science, Technology and a minor in psychology. Coursework included persuasion, entrepreneurship, computer science and advanced software engineering. Materials are available on my page.
- Extensive teaching and mentoring experience: Teaching and mentoring junior-mid level engineers. • Previous companies founded.
- Software consultants for some of the world's top 200 tech companies.